

Uses



Diamond blade saw

This sharp blade is coated with tiny diamonds.

This carbon filter absorbs harmful chemicals in water.



Water purifier filter

This battery has a graphite core that carries electric current.



Battery



Sturdy and lightweight frame

Carbon-fibre bicycle

This **strong** but **light** frame is made by fusing together carbon fibres.

This wheel is composed of carbon fibre and some other materials, which make it rigid.

Soft graphite leaves a mark when pressed against paper.



Pencil "lead" containing graphite, not lead

This flame is fuelled by carbon-rich oil.

Kerosene lamp



Polythene bag

This flexible plastic material is composed of chains of carbon and hydrogen atoms.



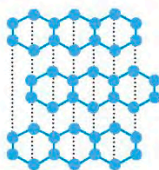
HARD AND SOFT CARBON

Diamond and graphite have different properties because of the arrangement of their atoms – tetrahedrons in diamond, and sheets of hexagons in graphite.



Hard as diamond

The tetrahedron, or pyramid structure, of atoms creates a rigid shape that is equally strong in all directions.



Soft as graphite

Graphite contains layers of carbon atoms that slide over each other easily, as there are only weak forces between them.

can cut into anything. Only a diamond can cut another diamond. Graphite is much softer, which is why it is used in **pencil "lead"**. It is also used in some batteries. Coal is currently the largest source of fuel for the generation of electricity, but its fumes are also known

to have harmful environmental and health effects. Crude oil, natural gas, and **coal** are hydrocarbons (compounds containing only hydrogen and carbon) that occur in nature. They can be used as fuels and as plastics for objects such as **polythene bags**.